STELLAR HACKATHON 2026 // PROBLEM STATEMENT #05

FinShield.

Master Engineering, Architecture & Decisions Document

Real-Time Multi-Signal Fraud Defense & Entity Resolution Platform for Digital Payments.
Production-Grade Blueprint: 5-Signal Convex Fusion, Live Cashfree PG Webhook Ingestion, and Post-Hoc Explainability.

XGBOOST 0.9709 ROC | 0.8418 PR-AUC | REAL KAGGLE ULB 284,807 | CASHFREE PG LIVE | FASTAPI | REACT 19 | NETWORKX | SHAP | 208 TESTS PASSING

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1. Executive Summary & Problem Statement Alignment

The Problem (Stellar Hackathon PS #05)

Digital payments—particularly instant rails like Unified Payments Interface (UPI), Instant SEPA, and Real-Time Gross Settlement—settle irrevocably in milliseconds. Traditional fraud detection architectures suffer from critical structural deficiencies: 1. **The Black-Box Dilemma**: Complex deep learning models produce uncalibrated probabilities without auditable rationales, violating financial compliance standards (e.g., RBI fraud reporting norms, FCRA, EU AI Act). 2. **The High False-Positive Curse**: Standard anomaly detectors or isolated classifiers trigger catastrophic false-positive spikes when legitimate users transact during travel, festivals, or high-value purchases, leading to cart abandonment and user churn. 3. **Syndicate & Device Blindness**: Per-transaction classifiers evaluate transactions in isolation, missing coordinated mule account rings, emulated rooted devices, and shared SIM hardware fingerprints. 4. **LLM Hallucination in Risk Scoring**: Autonomous generative agents plugged directly into financial decision loops introduce severe non-determinism, unpredictable risk drifts, and vulnerability to prompt injection attacks.

The FinShield Solution

FinShield is a **multi-signal, hybrid fraud intelligence and forensic platform**. It rejects single-point failure architectures in favor of **orthogonal signal fusion**: - **Zero Hallucination Risk**: The engine uses a calibrated, deterministic **Risk Fusion Engine** that fuses 5 independent vectors. - **Explainability by Design**: SHAP (SHapley Additive exPlanations) extracts exact mathematical feature contributions. An instruction-tuned **SLM (Small Language Model - Qwen2.5-0.5B)** articulates structured evidence for human forensic analysts, but is **strictly prohibited from calculating or overriding the numerical risk score**. - **Real-World Live Ingestion**: Fully integrated with live Cashfree Payment Gateway webhooks, processing real rupee transactions over public tunnels with sub-100ms end-to-end evaluation.

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2. End-to-End System Architecture

FinShield is intentionally architected as two cleanly decoupled repositories: 1. `Finsheild` **(ML Core Research & Training Engine)**: Houses dataset pipelines, feature extraction, model registries, training loops, SHAP explainers, synthetic stress suites, and QLoRA fine-tuning pipelines. 2. `Finsheild-App` **(FastAPI Backend + React Frontend)**: Serves as the real-time operational Command Center and Forensic Investigation Workstation.

Figure 1: FinShield End-to-End Multi-Signal Architecture & Forensic Audit Lifecycle

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3. The 5-Signal Fusion Engine — Mathematical & Algorithmic Deep Dive

Mathematical Formulation

Let a transaction be represented by a feature vector $\mathbf{x} \in \mathbb{R}^{36}$. The final risk score `R(x) \in [0, 1]` is computed as a linear convex combination of 5 normalized, bounded signal scores:

$$R(\mathbf{x}) = w_1 S_{\text{xgb}}(\mathbf{x}) + w_2 S_{\text{iforest}}(\mathbf{x}) + w_3 S_{\text{rules}}(\mathbf{x}) + w_4 S_{\text{behavior}}(\mathbf{x}) + w_5 S_{\text{graph}}(\mathbf{x})$$

Where the weights `\mathbf{w} = [0.35, 0.20, 0.20, 0.15, 0.10]` satisfy $\sum w_i = 1.0$.

Signal 1: Supervised Gradient Boosting (`XGBoost` — Weight: 0.35)

- **Role**: Detects non-linear tabular feature interactions learned from historical fraud patterns.
- **Model Configuration**:
- `n_estimators`: 500 trees
- `max_depth`: 5 (prevents overfitting to memorized card numbers)
- `learning_rate`: 0.05

- `scale_pos_weight`: Set to `\frac{N_{\text{legit}}}{N_{\text{fraud}}} ≈ 577.8` during training to counter extreme class imbalance (`0.17\%` fraud).
- Early stopping on Precision-Recall AUC (PR-AUC) using validation split.
- **Why It Matters**: Captures multidimensional correlations (e.g., amount + time-of-day + velocity + merchant category) that simple threshold rules miss entirely.

Signal 2: Unsupervised Anomaly Isolation (`Isolation Forest` — Weight: 0.20)

- **Role**: Zero-day fraud detection and novel anomaly isolation without relying on past labels.
- **Mathematical Principle**: Based on the insight that anomalies are few and structurally different, requiring fewer random axis-aligned splits to isolate in feature space.
- **Model Configuration**:
- `contamination`: 0.05
- **Critical Architectural Safeguard**: The Isolation Forest is fitted **strictly on legitimate transactions** (`y = 0`). This ensures the model constructs an unpolluted baseline profile of normal human spending behavior.
- **Anomaly Score Conversion**:

```
S_{\text{iforest}} = \frac{1}{1 + e^{k \cdot (\text{raw\_score} - t)}}
```

Maps the negative decision function into a normalized probability in `[0, 1]`.

Signal 3: Deterministic Rule Engine (Weight: 0.20)

- **Role**: Instant short-circuiting of non-negotiable risk violations and regulatory thresholds.
- **The 8 Deterministic Production Rules**:
- `HIGH_VELOCITY`: `≥ 5` transactions initiated within 5 minutes from the same account.
- `BURST_VELOCITY`: `≥ 8` transactions within 5 minutes (characteristic of automated brute-force card testing scripts).
- `NEW_DEVICE_HIGH_VALUE`: Transaction value `> \`500` (or `> ₹25,000`) initiated from a newly registered hardware ID (`t_{\text{registered}} < 24\text{h}$).
- `UNUSUAL_AMOUNT_ZSCORE`: Amount deviates by `> 3\sigma` from user historical rolling mean.
- `OFFHOURS_HIGH_VALUE`: High-value transaction (`> \`1,000` or `> ₹50,000$) initiated between 01:00 AM and 05:00 AM local time.
- `SHARED_DEVICE_RING`: Hardware device ID linked to `≥ 2` distinct user identities within 48 hours.
- `LOCATION_ANOMALY`: Geodesic distance from primary residence `> 300\text{km}` without preceding travel velocity buffer.
- `RAPID_COUNTRY_SWITCH`: Successive transactions from distinct countries within physically impossible flight travel windows.

Signal 4: Behavioral Profiling & Z-Score Drift (Weight: 0.15)

- **Role**: Personalizes risk evaluation to individual user spending habits.

- **Formulation**: For user u with rolling historical transaction mean μ_u and standard deviation σ_u:

$$z = \frac{x_{\text{amount}} - \mu_u}{\max(\sigma_u, \epsilon)}$$

$$S_{\text{behavior}} = \min\left(1.0, \frac{\max(0, z - 2.0)}{4.0}\right)$$

- **Why It Matters**: Prevents false positives. A ₹50,000 transaction from a high-net-worth individual with $\mu = ₹45,000$ yields $z \approx 0.2$ (zero risk), whereas the same transaction for a user with $\mu = ₹1,200$ yields $z = 18.2$ (maximum behavioral alert).

Signal 5: Entity Relational Graph (`NetworkX` — Weight: 0.10)

- **Role**: Discovers syndicated fraud rings, shared emulator devices, and mule account clusters.
- **Graph Topology**:
- **Bipartite Node Types:** `User`, `Account`, `Device`, `Gateway`, `Merchant`.
- **Edge Semantics:** `OWNS`, `AUTHENTICATES`, `SIGNS_PAYLOAD`, `ROUTES_VIA`, `CREDITS_ESCROW`.
- **Graph Metrics Evaluated:**
- **Shared Device Degree:** Count of distinct accounts originating from the same hardware fingerprint.
- **Mule Chain Depth**: Shortest path distance between an incoming transaction and known blacklisted liquidity cashout nodes.
- **Bipartite Co-Usage Cliques**: Cliques of size $k \ge 3$ trigger `SHARED_DEVICE_CLIQUE` alert ($S_{\text{graph}} \ge 0.85$).

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4. Every Technology & Library Decision — Why Chosen vs Why Rejected

Library / Tool	Chosen For	Rejected Alternatives & Engineering Rationale
`xgboost`	Superior tabular performance, built-in missing value handling, fast C++ tree construction, native early stopping on PR-AUC.	**Deep Neural Networks (MLP/ Transformer)**: Overfit tabular data, lack exact tree explainability, 50× higher latency. **RandomForest**: Slower inference, lacks gradient-directed split optimization.
`scikit-learn`	`IsolationForest` for unsupervised anomaly detection, `StandardScaler` for zero-leakage transforms, robust metric suites (`precision_recall_curve`, `roc_auc_score`).	**PyOD**: Heavier dependency footprint with redundant wrappers. **Custom Anomaly Code**: Unvalidated math; scikit-learn's Cython implementation is thoroughly audited.
`networkx`	Pure Python, lightweight in-memory graph representation, instantaneous BFS/DFS clique analysis, zero external daemon requirements.	**Neo4j / Memgraph**: Massive JVM/ Docker overhead, complex Cypher maintenance, overkill for sub-millisecond per-transaction ego-network queries.
`shap`	`TreeExplainer` calculates mathematically exact Shapley values in polynomial time (`O(TLD^2)`), satisfying regulatory proof of attribution.	**LIME**: Perturbation-based approximation produces non-deterministic explanations. **Feature Importances**: Global only; cannot explain why this specific transaction was blocked.
`Qwen2.5-0.5B-Instruct`	Ultra-compact (0.5B parameters), runs on CPU or <1GB VRAM, highly capable at JSON structured evidence reasoning, sub-100ms inference.	**GPT-4 / Claude / Llama-3-70B**: Astronomical cloud latency (1.5s–3s), recurring token billing, external data leak risks, massive compute requirements.
`peft` + `bitsandbytes` + `trl`	QLoRA (4-bit NF4 quantization, `r=8, \alpha=16`) enables fine-tuning the SLM on a standard Google Colab T4 GPU in <15 minutes.	**Full Parameter Fine-Tuning**: Requires high-end A100 GPUs, prone to catastrophic forgetting of grammar and conversational instruction following.
`FastAPI` + `Pydantic v2`	Asynchronous ASGI request handling, sub-millisecond response latency, automatic OpenAPI documentation, compile-time schema validation.	**Flask**: Synchronous blocking architecture, lacks built-in serialization and typed data contracts. **Django**: Enormous monolithic bloat unsuitable for microsecond microservice scoring.
`React 19` + `TypeScript`	Strict end-to-end typing, deterministic component state, virtual DOM reconciliation for high-speed streaming tables.	**Vanilla JS / jQuery**: Unmaintainable spaghetti state across 5 complex screens. **Angular**: Heavy boilerplate and rigid opinionated conventions.
`Tailwind CSS 4`	Zero-runtime CSS compilation, precise fintech aesthetic tokens, high-density forensic information architecture.	**Material UI / Bootstrap**: Generic consumer look, heavy CSS-in-JS runtime penalty, sluggish on large streaming tables.

Library / Tool	Chosen For	Rejected Alternatives & Engineering Rationale
`Cashfree Payments SDK`	Real enterprise Indian payment rail, UPI QR/Intent flows, comprehensive webhook event schemas (`PAYMENT_SUCCESS_WEBHOOK`).	**Razorpay / Stripe**: Requires business registration / GST for live sandbox webhooks; Cashfree provided instant testbed links for ₹1 and ₹1,00,000 real flows.
`cloudflared`	Creates zero-configuration encrypted tunnels from localhost to public edge, allowing live Cashfree webhooks to reach local dev engines.	**ngrok**: Ephemeral URLs change on every restart, strict bandwidth rate limits on free tiers.

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5. Empirical Benchmarks & Dataset Science

Benchmark Source of Truth: Real ULB Kaggle Credit Card Dataset

To maintain absolute empirical rigor and avoid "demo fiction", FinShield was trained and benchmarked on the gold-standard **ULB Kaggle Credit Card Fraud Dataset**: - **Total Transactions**: `284,807` - **Legitimate Transactions**: `284,315` (`99.828\%`) - **Fraudulent Transactions**: `492` (`0.172\%`) - **Feature Space**: 28 PCA-transformed features (`V_1 - V_{28}`) + Time + Amount. - **Stratified Split**: 70% Train (`n=199,364`), 15% Validation (`n=42,721`), 15% Test (`n=42,722`).

Rigorous Evaluation Results (Held-Out Test Set `N = 42,722`)

Model / Pipeline	ROC-AUC	PR-AUC	F1-Score	Precision	Recall	False Positives	False Negatives
XGBoost (500 Trees)	**0.9709**	**0.8418**	**0.8467**	**0.9206**	**0.7838**	**5**	**16**
Logistic Regression Baseline	0.9495	0.7005	0.7407	0.8125	0.6806	15	24

> **IMPORTANT NOTICE**
> **Production Context on False Positives**: Out of **42,648 legitimate transactions** in the test set, the XGBoost engine generated **only 5 false alarms** (`0.0117\%` false positive rate). In production banking, this translates to virtually zero friction for legitimate customers while capturing nearly 80% of fraud before settlement.

The Synthetic Stress Test Suite (Adversarial Progression)

Real-world fraud datasets often exhibit clear boundaries. To test how the model behaves under adversarial pressure, we designed 3 synthetic stress scenarios with increasing difficulty:

Figure 2: Adversarial Stress Test Progression vs Ground-Truth ULB Benchmark

1. **Easy Synthetic (`11.5\%` Fraud)**: PR-AUC **0.959**. Fraud instances cluster cleanly far from normal spending.

2. **1% Diluted (`1.07\%` Fraud)**: PR-AUC **0.553**. Realistic class imbalance introduces natural precision decay.
3. **Hard Overlap (`1.10\%` Fraud)**: PR-AUC **0.373**. Multimodal Gaussian clusters deliberately overlap legitimate user behaviors (e.g., fraud amounts match grocery bills, location jumps within normal transit times). This proved that **single models fail under adversarial conditions**, directly justifying the need for the **5-Signal Fusion Engine**.

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6. Live Real-World Payment Gateway Ingestion (Cashfree Integration)

Rather than merely generating synthetic data, FinShield was connected to **Cashfree Payments**, a Tier-1 Indian payment aggregator.

Architecture of Live Ingestion

1. **Public Tunnel**: `cloudflared tunnel --url http://127.0.0.1:8000` created a live public endpoint: `https://anaheim-resistant-follow-insulation.trycloudflare.com/api/webhooks/cashfree`
2. **Webhook Parser (`backend/main.py`)**:
3. Ingests `PAYMENT_SUCCESS_WEBHOOK` events.
4. Extracts nested fields: `data.order.order_amount`, `data.payment.cf_payment_id`, `data.payment.payment_method.upi`, `data.customer_details.customer_phone`.
5. Derives salted identity tokens and constructs transaction telemetry.
6. **Live Scoring Pipeline**: Passes the telemetry directly to `RealMLAdapter` (XGBoost + Scaler + Rules + Graph).

Real Live Payments Executed & Verified

Parameter	Live Payment 1 (Micro-Payment)	Live Payment 2 (High-Value Anomaly)
Payment Link	`code=fav8f4pqom50_AAAAAACpSlE`	`code=oav8fob7um50_AAAAAACpSlE`
Amount	**₹1.00**	**₹100,000.00 (₹1 Lakh)**
Transaction ID	`CF-216655019298976`	`CF-216656422682784`
Calculated Risk Score	0.035	0.890
Assigned Risk Tier	`LOW`	`CRITICAL`
Policy Decision	`APPROVE`	`BLOCK`
Rules Triggered	None (Clean baseline)	`BURST_VELOCITY`, `NEW_DEVICE_HIGH_VALUE`, `UNUSUAL_AMOUNT`, `UPI_HIGH_VALUE_ANOMALY`, `EXCEEDS_SINGLE_TRANSACTION_LIMIT`
Source Provenance	`LIVE_MODEL`	`LIVE_MODEL`

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7. Multi-Hop Entity Relational Graph & Forensic Workstation

The Visual & Structural Upgrade

In digital payment fraud, attacks are rarely isolated transactions; they are syndicates. We built the **Multi-Hop Entity Relational Canvas** in `frontend/src/pages.tsx` to replace static, clipped boxes with an interactive, publication-grade forensic graph.

4-Column Deterministic Topological Layout

Nodes are organized in a clean left-to-right causal transaction flow: 1. **Column 1 — Identity & Funding (`x = 115`)**: Payer Identity (`USER-8926`) and Funding Accounts / VPAs (`user-8926@hdfcbank`). 2. **Column 2 — Hardware & Origin (`x = 350`)**: Registered Hardware (`DEV-PIXEL-BASE`) vs Rogue Ingress Hardware (`DEV-NEW-CF`). 3. **Column 3 — Ingress Hub (`x = 580`)**: The Target Transaction (`₹100,000.00`), highlighted with pulsing risk borders. 4. **Column 4 — Switching & Settlement (`x = 810`)**: Gateway Switch (`Cashfree Switch`) and Beneficiary Escrow (`Merchant Escrow`).

Smooth Bézier Curves & Interactive Forensics

- **Curved Directed Edges**: Smooth cubic Bézier paths (`M x1 y1 C cx1 cy1, cx2 cy2, x2 y2`) connect card ports with zero cross-node line piercing.
- **Edge Badges**: Directed relationship labels (`OWNS_ACCOUNT`, `PAIRED_BASELINE`, `ROGUE_SIGNATURE`, `ROUTED_VIA`, `CREDITS_ESCROW`).
- **Anomalous Link Highlights**: Dashed crimson lines with drop-shadow glow filters highlight rogue hardware pairing and hardware mismatches.
- **Interactive Entity Inspector Drawer**: Clicking any node in the SVG canvas immediately opens a detailed forensic card displaying hardware fingerprints, velocity counts, trust scores, and cluster connectivity.

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8. Privacy, Compliance & Identity Tokenization

Zero-PII Guarantee in ML Training

In compliance with the **Digital Personal Data Protection (DPDP) Act 2023** and **GDPR**, raw Personally Identifiable Information (PII)—including phone numbers, Aadhaar numbers, PAN cards, and bank account numbers—**never touches feature matrices or model weights**.

Salted Cryptographic Hash Tokenization

- **Algorithm**: Salted SHA-256 with an isolated environment salt key:

```
\text{Token}(u) = \text{SHA-256}(\text{salt} \parallel u)
```

- **Masked Presentation**: Phone numbers are masked (`••••••••42`), documents are tokenized, and only cryptographic tokens are passed to the graph and feature pipeline.

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9. AI Forensic Copilot — Explainability Without Autonomous Risk

The Core Safety Principle

CRITICAL GUARDRAIL

LLMs must NEVER compute, calibrate, or override numerical fraud risk scores. In FinShield, the numerical risk score is computed solely by the deterministic 5-Signal Fusion Engine. The AI Forensic Copilot's sole mandate is to **translate complex mathematical evidence into plain, human-readable forensic rationales** for bank fraud analysts.

Model & Instruction-Tuning Setup

- **Base Model:** `Qwen/Qwen2.5-0.5B-Instruct`
- **Fine-Tuning Method**: QLoRA (`r=8, \alpha=16`, dropout `0.05`) via `peft` and `trl.SFTTrainer`.
- **Instruction Dataset**: Grounded JSON evidence templates generated from SHAP attribution bars, rule hits, and behavioral deviations.
- **Inference Speed**: ~80ms on GPU, ~300ms on CPU, completely eliminating multi-second cloud API calls.

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10. Frontend Ergonomics & Design System

The FinShield UI was built to eliminate the clutter of traditional enterprise banking software while maximizing information density.

Typography Hierarchy

- **Headings & Display**: `Plus Jakarta Sans` (800/700 weight, tight tracking `-0.035em` to `-0.04em`) — commanding fintech presence.
- **Body Prose**: `Inter` (with optical kerning `cv02`, `cv03`, `cv04`, `cv11`) — ultra-crisp readability.
- **Telemetry, Hashes & Figures**: `JetBrains Mono` (with tabular figures `font-feature-settings: "tnum" 1, "zero" 1`) — perfect vertical digit alignment.
- **Accents & Quotations**: `Space Grotesk` (geometric accents).

Editorial Color Palette

- **Canvas / Background**: `#F2EFE7` (warm editorial off-white)
- **Primary Ink**: `#171916` (deep charcoal carbon)
- **FinShield Brand Neon**: `#FF5B35` (high-visibility safety orange/crimson)
- **Safe / Approved Green**: `#2E7D32`
- **Surface Elevation**: `#E7E4DB`

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11. 10-Step Evaluator & Judge Verification Walkthrough

Evaluators can verify the entire platform end-to-end using this structured 10-step sequence:

1. **Step 1: Check System Health & Provenance**:
2. Navigate to `/api/health`.
3. Verify `adapter: "real"`, and confirm that XGBoost, Scaler, SHAP, and Graph are loaded live.
4. **Step 2: Inspect Real ULB Kaggle Metrics**:
5. Navigate to the **Model Performance** screen (`/#/performance`).
6. Confirm the held-out test ROC-AUC is **0.9709**, PR-AUC is **0.8418**, and only 5 false alarms occurred across 42,722 test samples.
7. **Step 3: Review Adversarial Stress Tests**:
8. Compare the synthetic experiment progression (Easy Synthetic 0.959 → Diluted 0.553 → Hard Overlap 0.373).
9. **Step 4: Live Cashfree Payment Ingestion (₹1 Micro-Payment)**:
10. Locate transaction `CF-216655019298976` in the Command Center table.
11. Confirm that the score is **0.035**, risk tier is `LOW`, and policy decision is `APPROVE`.
12. **Step 5: Live Cashfree Anomaly Block (₹1,00,000 High-Value Payment)**:
13. Locate transaction `CF-216656422682784` in the Command Center table.
14. Confirm that the score is **0.890**, risk tier is `CRITICAL`, and policy decision is `BLOCK`.
15. **Step 6: Forensic Deep Dive (5-Signal Radar)**:
16. Click "Investigate" on `CF-216656422682784` to open `/#/investigate/CF-216656422682784`.
17. Verify the 5-signal breakdown: XGBoost (0.00), Anomaly (0.75), Rules (`BURST_VELOCITY`, `NEW_DEVICE_HIGH_VALUE`), Behavioral (0.62), Graph (0.42).
18. **Step 7: SHAP Feature Attribution**:
19. Inspect the horizontal waterfall bars. Note how `amount`, `velocity`, and `new_device` push the decision toward fraud, while `merchant_category` acts as a dampener.
20. **Step 8: Interactive Multi-Hop Entity Relational Canvas**:
21. Examine the 4-column layout on the investigation page.
22. Click on `DEV-NEW-CF` to view the Entity Inspector Drawer. Notice the `UNRECOGNIZED_FINGERPRINT` and velocity burst indicators.
23. Click the "Anomalous Paths Only" toggle to filter out noise and isolate the rogue attack route.
24. **Step 9: AI Forensic Copilot Rationale**:
25. Click "Generate Copilot Explanation".
26. Confirm the model articulates the exact rule violations and behavioral drifts into a structured narrative without modifying the score.
27. **Step 10: Privacy & Identity Tokenization**:
    - Navigate to `/#/privacy/USER-8926`.
    - Confirm the salted SHA-256 token, phone masking, and zero-PII compliance badges.

12. Repository Layout, Tooling & Deployment Matrix

```
Finsheild-Ecosystem/
├── Finsheild/                      # ML Core Research & Training Engine
│   ├── src/finsheild/
│   │   ├── data/                   # Kaggle loader & stratified splits
│   │   ├── model.py                # XGBoost, LightGBM, LogReg registries
│   │   ├── train.py                # Training loops with early stopping
│   │   ├── inference.py            # Low-latency inference pipeline
│   │   ├── features/               # 36 leakage-safe feature definitions
│   │   ├── behavioral/             # Online rolling user profiles
│   │   ├── anomaly/                # IsolationForest (legit-only fitting)
│   │   ├── rules/                  # 8 production deterministic rules
│   │   ├── graph/                  # NetworkX multi-hop entity resolution
│   │   ├── risk_fusion/            # 5-signal linear convex combination
│   │   ├── explain/                # SHAP TreeExplainer
│   │   └── finetune/               # QLoRA Qwen2.5-0.5B instruction tuning
│   ├── models/                     # Serialized weights (XGBoost, Scaler)
│   ├── evaluation/reports/         # Benchmark JSON & Markdown reports
│   └── tests/                      # 208 passing unit & integration tests
│
└── Finsheild-App/                  # Presentation & Workstation Engine
    ├── backend/
    │   ├── main.py                 # FastAPI app (13 REST endpoints + webhooks)
    │   ├── schemas.py              # Pydantic data contracts
    │   ├── metrics_loader.py       # Real ULB benchmark ingestion
    │   ├── services/store.py       # In-memory store & graph generator
    │   └── adapters/
    │       ├── real_adapter.py     # Connects to live Finsheild ML Core
    │       └── mock_adapter.py     # Transparent fallback with honesty labels
    ├── frontend/
    │   ├── src/pages.tsx           # Command Center, Investigation, Performance, Graph
    │   ├── src/api.ts              # Typed API client
    │   └── src/index.css           # Editorial tokens & typography
    └── start.sh                    # One-click dual server launcher
```

Remote Deployments & Public Access Links

- **Hugging Face Space**: https://huggingface.co/spaces/shaikhakramshakil/Finsheild
- **Direct Static Application URL**: https://shaikhakramshakil-finsheild.static.hf.space
- **GitHub Application Repo**: https://github.com/shaikhakramshakil/Finsheild-App
- **GitHub ML Core Engine Repo**: https://github.com/shaikhakramshakil/Finsheild
- **Live Cashfree Webhook Ingress URL**:
  `https://anaheim-resistant-follow-insulation.trycloudflare.com/api/webhooks/cashfree`

FinShield Fraud Defense Platform — Engineering Rigor. Data Honesty. Multi-Signal Fusion.